
Feedback prize 2021 - Evaluate Student Writing

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Abstract

This report analyzes the methodologies and outcomes of the Feedback Prize - Evaluating Student Writing competition hosted on Kaggle [7]. The competition aimed to advance the development of automated systems for automatically segmenting and classifying argumentative elements in student essays from grades 6 to 12. This initiative aimed to enhance Assisted Writing Feedback Tools (AWFTs), providing educators with efficient means to assess and support student writing. The findings underscore the efficacy of pretrained transformer-based models in automating the evaluation of student writing and highlight the importance of continued research in this domain to support educators and enhance student learning outcomes.

1 Introduction

1.1 Automated Writing Feedback

Less than a third of high school seniors are proficient writers, according to the National Assessment of Educational Progress. One way to help students improve their writing is via automated feedback tools, which evaluate student writing and provide personalized feedback.

There are currently numerous automated writing feedback tools, but they all have limitations. Many often fail to identify writing structures, such as thesis statements and support for claims, in essays or do not do so thoroughly. Additionally, the majority of the available tools are proprietary, with algorithms and feature claims that cannot be independently backed up. More importantly, many of these writing tools are inaccessible to educators because of their cost. This problem is compounded for under-serviced schools that serve a disproportionate number of students of color and from low-income backgrounds. In short, the field of automated writing feedback is ripe for innovation that could help democratize education.

1.2 Organizers

Georgia State University (GSU) is an undergraduate and graduate urban public research institution in Atlanta. U.S. News and World Report ranked GSU as one of the most innovative universities in the nation. GSU awards more bachelor's degrees to African-Americans than any other non-profit college or university in the country. GSU and The Learning Agency Lab, an independent nonprofit based in Arizona, are focused on developing science of learning-based tools and programs for social good.

1.3 Task and Contributions

The objective of this work is to automatically segment texts and classify argumentative and rhetorical elements in essays such as Lead, Position, Claim, Counterclaim, Rebuttal, Evidence, and Concluding Statement written by 6th- 12th-grade students. Test data contained 10000 unseen essays. While the public leaderboard shows results of 39% of test data, the private leaderboard shows results of the remaining 61% test data.

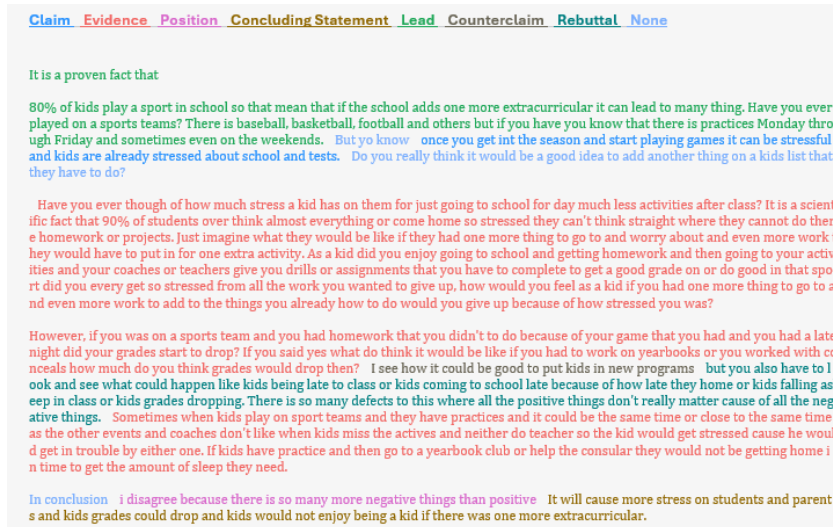


Figure 1: Sample essay highlighted with different color for each discourse type

I have used several Python plotting libraries, such as matplotlib, seaborn, and plotly, to analyze and visualize the dataset and understand the properties of the training dataset and essay elements to develop an appropriate model pipeline. I have leveraged transformer-based architectures, notably fine-tuning pre-trained models like BERT [5] to achieve high accuracy in segmenting essays. These models demonstrated the potential of large language models in educational applications, achieving performance levels approaching human annotator agreement. Qualitative analysis is done to understand what the network is learning and how it's treating each data sample at an embedding level. Several post-processing methods were proposed to further improve performance. The overall contributions of this work are:

- Performed Data Analysis and Visualization of essay dataset using various python libraries
- Utilized pretrained transformer models to fine-tune the essays dataset and develop a faster and cost-effective automated writing feedback tool.
- Qualitative Analysis of model's output, feature vectors and Attention weights through python plotting and machine learning techniques
- Proposed several post-processing techniques to further improve the accuracy of the segmented essay by a fine-tuned model

2 Data Analysis

Organizers of the competition have provided a set of text files containing an essay from each student. Each element of the essay is called a discourse, and there are **Claim**, **Evidence**, **Position**, **Concluding Statement**, **Lead**, **Counterclaim**, and **Rebuttal**. As part of the training labels, a csv file is provided, which has information about segments from an essay, its start and end positions in the essay, the type of discourse, and its count if there are multiple segments of the same discourse type. There are some essays where some part of the text is not labelled, which is given a label **None** in figures. The following figure 1 is a sample essay with text highlighted with different colors for each discourse type.

To understand the distribution of each discourse type, a histogram of tokens is plotted across all the training essays in the figure 2 and it shows a clear imbalance in tokens where Claim and Evidence seem to be of majority, followed by unlabeled tokens. While the distribution of tokens gives the idea of balance in the tokens, understanding how and where each essay contains each discourse type's tokens is equally important. Left figure of 3 shows the distribution of the fraction of each discourse type tokens in the training essays. While Evidence seems to constitute most of the tokens, there are more outliers in Claim, which makes its overall token count higher. Right figure of 3 shows

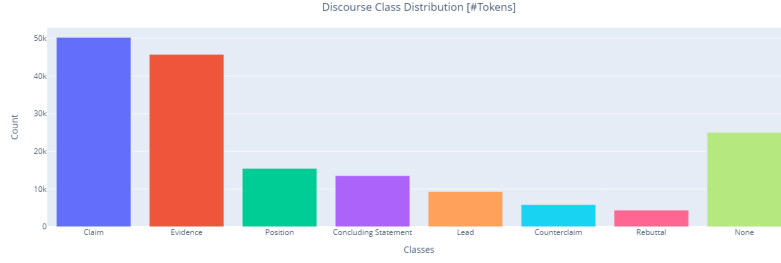


Figure 2: Token distribution of training data for different discourse types

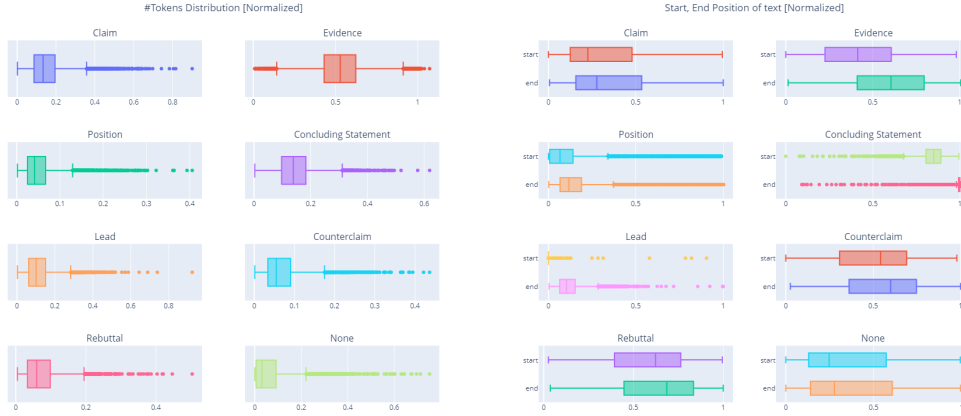


Figure 3: Left: % Distribution of number of discourse type tokens. Right: Start and End positions distribution of each discourse type segment in an essay.

distribution of start and end token positions of each discourse type in training essays. Most of the essays have *Lead* positioning at the beginning of the essay and *Concluding Statement* positioning at the end of the essay, which is grammatically right.

3 Methodology

This task can be considered as a token classification problem. Pretrained model's embeddings are passed through dropout layer to obtain more robust features. FCN layer is then added to further refine BERT embeddings to adapt to our task. A dropout layer is added to prevent overfitting. Finally a Linear layer with softmax activation is added to output classification probabilities for each token. Figure 4 shows the architecture implemented for this task.

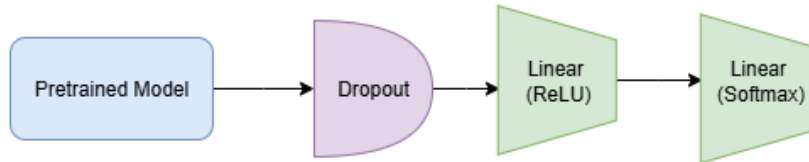


Figure 4: Model architecture

4 Experiments

4.1 Pretraining Models

As mentioned in the section above 3, we finetune *google-bert/bert-large-uncased* for this task. Several experiments are done by varying freezing layers, loss criterion during fine-tuning.

4.2 Data Preprocessing

All the models are trained using the following data splitting strategies:

- **Kfold**: Labelled data is split into 5 folds, and 5 models are trained with different folds
- **Stratified Kfold**: Each labeled sample is given a pseudo label based on the length of each discourse type in its essay, and then the above Kfold method is used but with stratification.

Stratified Labeling is done by fitting KMeans clustering on the lengths array of all discourse types, including *None*. There is also a **Train-Val** splitting strategy that can be tested if the computational resources are limited

4.3 Training Settings

All the models are trained on JarvisLabs.AI’s RTX6000Ada GPU with 48GB memory. AdamW optimizer with an initial learning rate of $2e-5$ is used along with MultiStepLR learning rate scheduler. All the Dataloader and Model pipelines are implemented using Pytorch and Pytorch Lightning. We use 512 as the sequence length with a batch size 16, trained for 10 epochs for all the experiments. Due to non-uniform class distribution, we also experimented with class weighting in the loss function. Classes are weighed based on the overall token count of each class in the entire training set.

4.4 Post-processing

Finetuned model results can be further improved by using the following post-processing methods.

- **Soft Voting**: Multiple finetuned models softmax probabilities are averaged for all tokens.
- **Hard Voting**: Multiple fine-tuned models outputs are combined through majority voting of predicted class for each token.
- **Span Average**: Each prediction’s start and end positions can be averaged across multiple fine-tuned models
- **Span Repair**: If, for a given essay, there are multiple prediction spans with the same discourse type consecutively with less gap, all those spans are combined to give one prediction output.

5 Results

5.1 Ablation Analysis

Table 1: Results of all the experiments.

Splitting strategy	Top k Encoder layers frozen	Class Weighting	Post-processing	Private score	public score
KFold	20	Disabled	Soft Voting	0.396	0.385
KFold	20	Disabled	Hard Voting	0.393	0.378
KFold	20	Disabled	Soft Voting + Repair Spans + Bound Average	0.396	0.385
Stratified KFold	20	Disbaled	Soft Voting + Repair Spans + Bound Average	0.393	0.388
Stratified KFold	20	Enabled	Soft Voting + Repair Spans + Bound Average	0.393	0.388
Stratified KFold	15	Enabled	Soft Voting + Repair Spans + Bound Average	0.507	0.499
Stratified KFold	10	Enabled	Soft Voting + Repair Spans + Bound Average	0.520	0.510

Table 1 shows the results of all the experiments. While Soft Voting has shown improvement over Hard voting, other post-processing methods didn’t have any affect on the performance. Even class

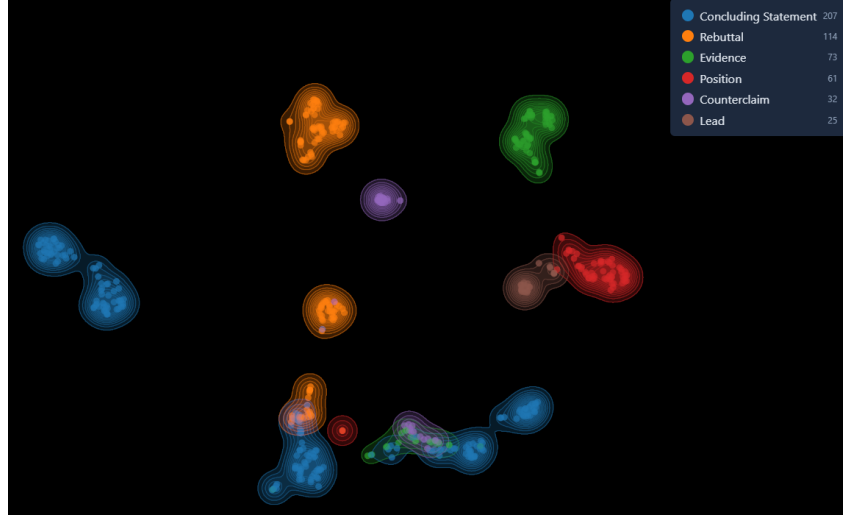


Figure 5: tSNE of Embeddings of last hidden state for one of the training data samples

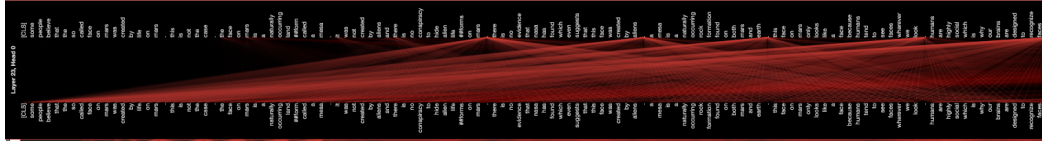


Figure 6: Attention visualization of one of the head's from last encoder layer

weighting has little or no affect on leaderboard score, while it has shown improvement in the Cross validation scores. Unfreezing encoder layers has shown significant improvement consistently.

While this report focusses only on BERT Large Uncased model, I also tested Funnel Transformer [4]. Longformer [2] and XLM RoBERTa [3] initially, which gave very good results (0.5x for all). But because of GPU resource constraints, I couldn't explore them extensively. Also, having a BiLSTM [6] in place of a Dense layer on top of BERT Large Uncased model has a significant improvement (was able to get to 0.51 with BERT Large Uncased and 20 frozen encoder layers). I believe adding all the novelties that were introduced in this work to all the above mentioned models will surely improve the overall leaderboard score further.

5.2 Qualitative Analysis

We used some existing explainability tools to understand what the network is learning and how to interpret them. Figure 5 shows tSNE plot of one of the training data sample's last hidden states. You can clearly see the clustering of different classes, such as Concluding Statement, Rebuttal, Lead, and Evidence. There is also some overlapping region of Counterclaim, Concluding Statement, and Evidence, which could be the confusing part of the essay to the model, and it probably made an error for those tokens. We took help of Apple's [1] Embedding Atlas to plot embeddings.

Interpreting Attention weights is one way to understand what a given Transformer model is learning. BertViz [8] is a famous python library that can show attention weights in different views. Figure 6 is a model view of one of the heads from the last encoder layer for one training data sample. All the words are attending to certain full stops which are the end of each essay type. For example first full stop is the end of the class Lead.

6 Summary

Acknowledgements

I would like to thank Kaggle and the organizers, Georgia State University and The Learning Agency Lab, for providing an invaluable opportunity to work on a real-world educational NLP problem. I am thankful for the high-quality dataset and the detailed competition framework that allowed exploring advanced machine learning techniques in the context of student writing assessment. Furthermore, I acknowledge the vibrant Kaggle community for sharing insightful discussions, kernels, and solution write-ups that greatly enriched our understanding and approach to the problem.

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